PLAIN-LANGUAGE BRIEF · OCTOBER 2026

A smarter way to plan lab experiments

Turning fish waste into antioxidant peptides - and using every experiment to choose the next one. This page explains what the project does, what the test found, and what it still cannot claim.

Real data
31 batches · 7 time-course curves

Synthetic test
100 unseen computer-generated worlds

Real campaign
0 of 15 runs measured so far

Why this project can be useful

Less wasted bench time
Each run is chosen to teach the most, instead of guessing.

Covers more ground
It explores enzyme, pH, temperature, time and dose combinations a person might miss.

Learns as it goes
Every measured batch updates the next three suggestions.

Keeps a paper trail
Every suggestion has a reason and a record, so the lab stays in control.

How the loop works

1. **Start with data** — 31 historical batches and 7 time curves.
2. **Predict** — A physics model estimates DH; a ranker scores antioxidant potential.
3. **Choose** — The planner proposes the next 3 runs.
4. **Test** — The lab measures the real result.
5. **Learn** — Update the model and repeat.

DH = how much protein is broken into small peptides (the main score). **Planner** = the model that chooses the next experiments. The planner is the part under test; the physics model is fitted to Alcalase only.

What the lab gets

- 3 suggested runs per round
- A reason for each suggestion
- A record of every update
- A model that improves with data

What it does not do

- Replace lab judgment
- Guarantee a better run
- Yet optimise antioxidant score or cost

What the 30-run budget means



15 shared seed runs · 15 adaptive runs chosen by each method

What the synthetic test found

100 unseen computer-generated worlds. Same 30-run budget for every method. Higher is better.



* Best possible is a cheat sheet that knows the synthetic truth. It is not a real method. Values are the best true DH reached by run 30.

The planner beat random, space-filling and grid search. A simple one-setting-at-a-time search did better: 40.5 vs 38.4. So this is promising, not proven.

What this means

Promising:
it beat three common designs at the same 30-run budget.

Not proven:
the simplest strong baseline still won.

Next:
test on the real campaign, not more synthetic worlds.

Computer-generated lab only
Not real lab proof. The test world is not the real process.

Not the best baseline
One-setting-at-a-time won 55.2% of worlds; the planner won 29.2%.

Planner ranking is weak
It wins mostly by where it looks, not by a smart prediction.

Real campaign missing
0 of 15 runs have returned. Real-data validation is future work.

Cold-start caveat
The tested variant was a cold-start version. Production keeps the safer rule; realistic history removes the advantage.

What would make it useful: run the 15-batch campaign and compare with one-setting-at-a-time on real data. If it holds, the lab can spend fewer runs finding good conditions. The planner is a decision aid, not a replacement for lab judgment. **Plain-English glossary:** DH = degree of hydrolysis; ORAC = antioxidant score; planner = experiment chooser; one-at-a-time = change one setting, measure, repeat.

Full report: `data/synthetic/benchmark_final_report.md`
Deeper story: `reports/twin_story.html`
No real-world trial-reduction claim.